Successive differences

Animath — Stage olympique de Sophia-Antipolis · 2006 · France

Four integers a, b, c, d, not all equal, are given. From the quadruple (a, b, c, d) we form (a', b', c', d') with $a' = a - b$, $b' = b - c$, $c' = c - d$ and $d' = d - a$. The operation is repeated from (a', b', c', d'), and so on, any number of times. Show that after a while one of the components of the quadruple obtained will be greater than 20062007.